

FINANCIAL AI AGENT

An Intelligent RAG-Based Financial Advisor for Local and Global Markets

Document 1: Project Planning & Management

Graduation Project — Digital Egypt Pioneers Initiative (DEPI)

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Table of Contents

1. Project Proposal

1.1 Overview

The Financial AI Agent is a comprehensive, AI-powered financial advisory platform designed to serve both the Egyptian local market and global financial sectors simultaneously. The system integrates real-time market data ingestion, a Retrieval-Augmented Generation (RAG) conversational engine, and a predictive machine-learning pipeline to give investors reliable, mathematically grounded insight instead of the speculative or hallucinated output typical of general-purpose chatbots.

The platform tracks global equities and currency pairs through the yfinance API and complements this with locally scraped data — most notably real-time Egyptian gold prices from the Isagha index — so that users receive a single, unified view across both markets. A large language model (Llama-3.3-70b-versatile, served through Groq's low-latency inference engine) answers user questions, but its responses are strictly grounded in retrieved, up-to-the-minute data stored in a PostgreSQL database extended with pgvector, rather than in the model's own memorized knowledge.

1.2 Objectives

- Absolute Accuracy — deliver real-time, verifiable price quotes for global and local instruments.
- Contextual Advice — generate investment insights that are tailored to current market conditions rather than generic commentary.
- Accessibility — provide an easy-to-use, Arabic-first conversational interface suited to the regional demographic, while correctly handling mixed Arabic/English (RTL/LTR) content.
- Predictive Edge — extend the platform beyond conversational reporting with institutional-grade machine-learning forecasts of statistically significant price movements.
- Trust & Safety — eliminate LLM hallucination by constraining every financial claim to retrieved, sourced data.

1.3 Scope

In Scope

- Real-time ingestion of global equity and currency data via the yfinance API.
- Automated web scraping of the Egyptian gold market (Isagha) using BeautifulSoup.
- A RAG pipeline combining a local embedding model (BAAI/bge-m3) with a PostgreSQL + pgvector knowledge store.
- A conversational Streamlit front end supporting Arabic and English.
- A supervised machine-learning pipeline (XGBoost / LightGBM ensemble) that forecasts 10-day statistically significant stock moves for a universe of 40 major US-listed tickers.
- Risk-adjusted backtesting of trading signals against a benchmark index (SPY).

Out of Scope (current release)

- Direct trade execution or brokerage integration.

- Coverage of the Egyptian Stock Exchange (EGX) and cryptocurrency markets (planned for a future phase).
- Automated push/email alerting (planned for a future phase).
- Portfolio-level optimization and rebalancing (planned for a future phase).

2. Project Plan

2.1 Timeline & Milestones (Gantt Overview)

The project followed the DEPI graduation-project schedule, structured into six sequential phases over roughly 22 weeks. The chart below summarizes phase duration; shaded cells indicate the active weeks for each phase.

Phase	Wk1-2	Wk3-4	Wk5-6	Wk7-8	Wk9-10	Wk11-12	Wk13-14	Wk15-16	Wk17-18	Wk19-20	Wk21-22
Project Planning & Management											
Literature Review & Requirements											
System Analysis & Design											
Data Ingestion & RAG Pipeline											
Predictive ML Pipeline											
Frontend & Integration											
Testing, Reporting &											

Phase	Wk1-2	Wk3-4	Wk5-6	Wk7-8	Wk9-10	Wk11-12	Wk13-14	Wk15-16	Wk17-18	Wk19-20	Wk21-22
Final Presentation											

2.2 Deliverables

Milestone	Deliverable(s)	Target Date
Planning & Management	Project Proposal, Plan, Roles, Risk Plan, KPIs	2/20/2026
Literature Review	Comparative analysis, feedback log, grading criteria	4/20/2026
Requirements Gathering	Stakeholder analysis, use cases, functional/non-functional requirements	4/20/2026
System Analysis & Design	Architecture, ERD, DFD, sequence/activity/state/class diagrams, UI mockups	5/1/2026
Implementation	Ingestion pipeline, RAG engine, ML pipeline, Streamlit app, source code repo	7/10/2026
Final Presentation & Testing	Test reports, user manual, technical docs, final slide deck, demo	7/17/2026

2.3 Resource Allocation

Team Member	Primary Responsibility	Effort Area
Ahmed Sayed	RAG pipeline & vector database	Backend / Data
Ahmed Medhat	LLM integration & prompt engineering	Backend / AI
Ahmed Mohamed	Data ingestion (APIs & scraping)	Data Engineering
Omar Waleed	Streamlit frontend & UX (RTL/LTR)	Frontend
Ahmed Mohamed Montaser	Predictive ML pipeline (XGBoost/LightGBM)	Data Science
Abdelrahman Tarek	Testing, backtesting & documentation	QA / Documentation

3. Task Assignment & Roles

Roles were assigned according to each member's primary skill area, with cross-review built into every phase so that no single component (data, AI, or frontend) was owned without a second reviewer.

Role	Responsibilities	Owner(s)
Data Engineer	Build and maintain the yfinance ingestion job and the BeautifulSoup gold-price scraper; ensure data quality checks (OHLC consistency, deduplication)	Ahmed Mohamed
RAG / Vector-DB Engineer	Design the PostgreSQL + pgvector schema, run local BAAI/bge-m3 embeddings, implement retrieval logic	Ahmed Sayed
AI / Prompt Engineer	Configure the Groq/Llama-3.3-70b integration, role-prompting, boundary setting, RTL/LTR formatting rules	Ahmed Medhat
Frontend Engineer	Build the Streamlit chat UI, session state, live-refresh control	Omar Waleed
Data Scientist	Build and validate the feature-engineered predictive model (financial_advisor.ipynb): feature set, targets, ensembling, backtesting	Ahmed Mohamed Montaser
QA / Documentation Lead	Test planning, embargoed cross-validation review, documentation and reporting	Abdelrahman Tarek

4. Risk Assessment & Mitigation Plan

Risk	Likelihood	Impact	Mitigation
Global data-provider downtime / missing endpoints for regional currencies (HTTP 404s)	Medium	High	Compute cross-rates mathematically from a reliable USD base rate instead of depending on a direct regional feed
LLM hallucination of financial figures	Medium	High	Strict RAG grounding; boundary-setting prompts that force refusal outside retrieved context
Data leakage across time in the ML pipeline	Medium	High	Embargoed, purged time-series cross-validation with an explicit 20-day gap between folds
Class imbalance in the 10-day target (buy signal $\approx$ 17% of samples)	High	Medium	scale_pos_weight tuning, precision-focused threshold selection, ensembling across seeds
Overfitting from hand-crafted interaction features	Medium	Medium	Feature-ablation test comparing full vs. reduced feature sets before finalizing
RTL/LTR rendering conflicts in the Arabic-first UI	Medium	Medium	Prompt-level bracketing of numerical values/tickers to isolate LTR runs inside RTL text
Schedule slippage across six team members	Medium	Medium	Fixed milestone dates aligned to the DEPI schedule with weekly progress check-ins
Model source availability across GPU/CPU inference (device mismatch warnings)	Low	Low	Explicit device placement checks during inference; fallback to DMatrix-based prediction

5. Key Performance Indicators (KPIs)

5.1 Product / System KPIs

KPI	Target	Notes
Data uptime for regional currency pairs	100%	Achieved via USD cross-rate math workaround
Response latency (chat query)	Near-instant (sub-second generation)	Enabled by Groq LPU inference of Llama-3.3-70b
System uptime	>99%	Streamlit app + PostgreSQL availability
User adoption / session depth	Multi-turn sessions retained	Session-state persistence across follow-up questions
Language coverage	Arabic + English, correctly rendered	RTL/LTR bracketing fix validated in UI

5.2 Predictive Model KPIs (from financial_advisor.ipynb)

Metric	Result	Context
Test AUC (final XGBoost ensemble)	0.6309	3-seed ensemble on held-out 2024+ test period
Test Average Precision	0.2979	vs. base rate of ~0.17 positive class
Backtest cumulative return (capped, high-precision threshold)	+296.9%	10-day hold, 10-bps round-trip cost, capital cap of 10 concurrent positions
Backtest Sharpe ratio (capped)	1.46	vs. SPY benchmark Sharpe of 1.20 over the same period
Backtest Sortino ratio (capped)	2.12	vs. SPY 1.56
Backtest Calmar ratio (capped)	9.29	vs. SPY 2.84
Win rate (high-precision signals)	65.18%	1,100 simulated trades in the test window
Maximum drawdown (capped)	31.97%	vs. SPY 18.90% over the same window

These KPIs were generated from the model-development notebook and are reported here for planning and success-tracking purposes; see the System Analysis & Design document for full methodology.